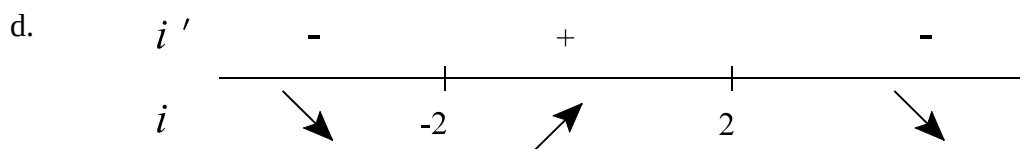


$$i(x) = \frac{x}{x^4 + 48}$$

a. Domain = $\mathbb{R}$

b. Intercepts (0,0)

c. $i'(x) = \frac{-3x^4 + 48}{(x^4 + 48)^2} \Rightarrow \text{C.N. } x = -2, 2$



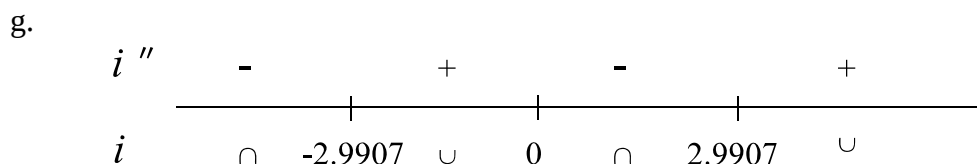
i is decreasing on $(-\infty, -2) \cup (2, \infty)$

i is increasing on $(-2, 2)$

e. $P(-2, -1/32)$ is a relative minimum on the graph

$P(2, 1/32)$ is a relative maximum on the graph

f. $i''(x) = \frac{12x^3(x^4 - 80)}{(x^4 + 48)^3} \Rightarrow 2^{\text{nd}} \text{ C.N. } x = -2\sqrt[4]{5}, 0, 2\sqrt[4]{5}$



i is concave up on $(-2\sqrt[4]{5}, 0) \cup (2\sqrt[4]{5}, \infty)$

i is concave down on $(-\infty, -2\sqrt[4]{5}) \cup (0, 2\sqrt[4]{5})$

h. Inflection points $P(-2.9907, -0.2336)$, $P(0, 0)$, $P(2.9907, 0.2336)$

i. -4, -0.01320

-3, -0.02330

-2, -0.03130

-1, -0.02040

0, 0.

1, 0.02040

2, 0.03130

3, 0.02330

4, 0.01320

